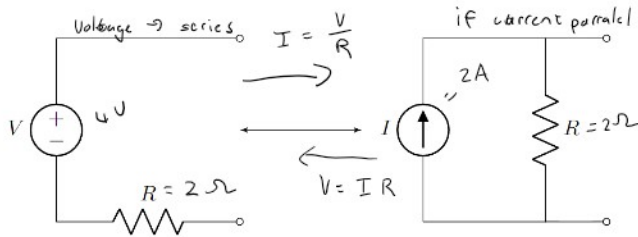


L14 - Source Transformation

Thursday, March 14, 2024 11:09 AM

Core Principles:



- $R_T = V_T / I_T$
- NEVER transform the measured/output/dependent values

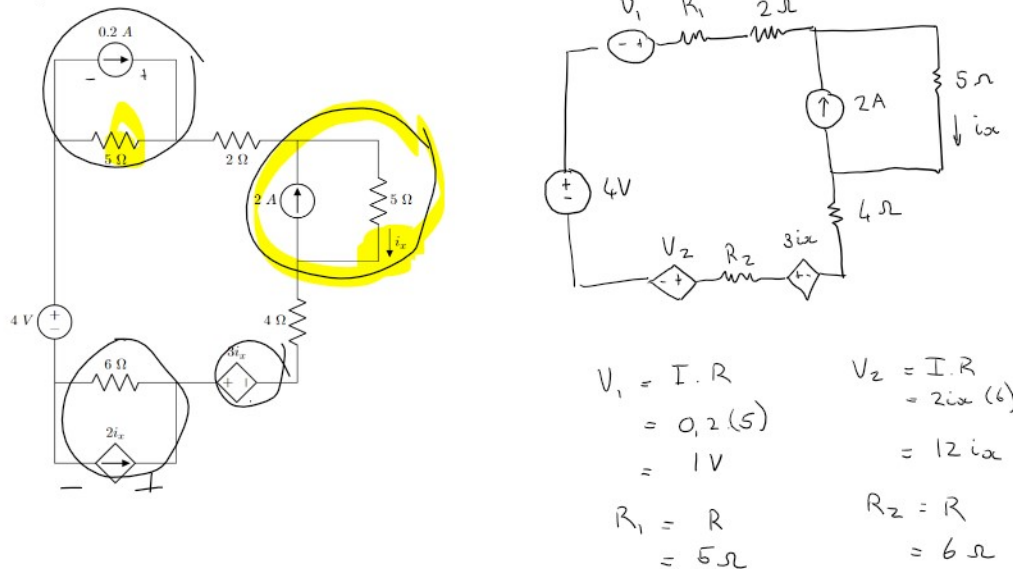


Why source transformation?

Remove loops or nodes that increase the number of equations

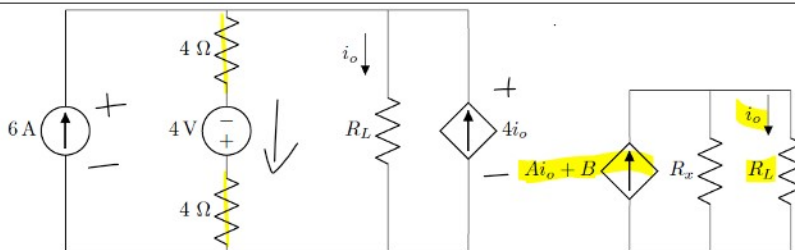
→ in eg.1, no transforms : 4 mesh equations or 6 node equations
with transformation 2 equations

Example 1

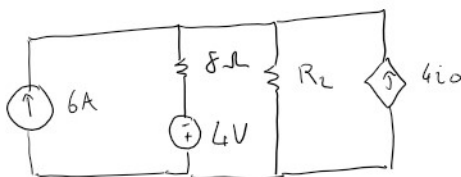


Example 2

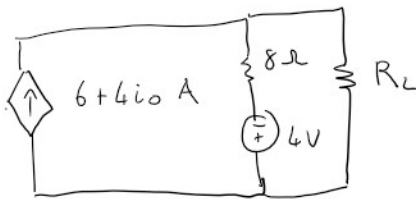
Simplify the circuit on the left by means of source transformations so that it looks like the circuit on the right. Determine R_x , A and B.



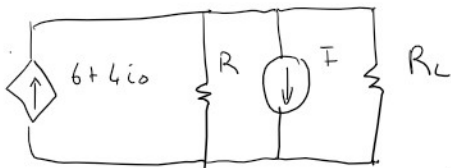
① Combine 4Ω resistors in series



② Combine current sources

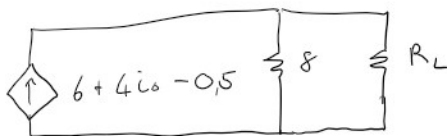


③ Source transform 8Ω & $4V$



$$R = 8\Omega \quad I = \frac{V}{R} = \frac{4}{8} = 0,4$$

④ combine current sources



$$\therefore R_{\alpha} = 8\Omega$$

$$A = 4$$

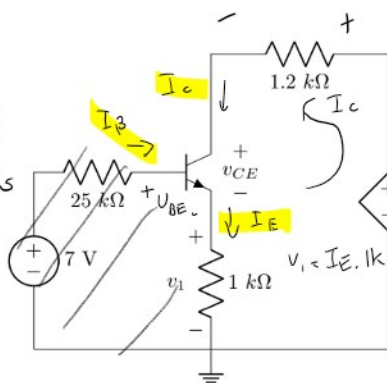
$$B = 5,5$$

Complete while getting seated:

If $i_B = 50 \mu A$, $v_{BE} = 0.7 V$ and $\beta = 99$, solve for v_{CE} .

Transistor

- 1) find I_B
- 2) use $I_B : I_C$ rel
- 3) mesh analysis



$$I_B \beta = I_C$$

$$I_E = I_B + I_C$$

$$I_E \approx (1 + \beta) I_B = (1 + 99) I_B = 100 I_B$$

KVL:

$$\begin{aligned} -3v_1 + I_C(1,2k) + v_{CE} + 1k(I_E) &= 0 \\ -3(1k(I_E) + I_C(1,2k) + v_{CE} + 1k(I_E) &= 0 \end{aligned}$$

09 $v_{CE} [2]$ find I_B

$$-3(1k)(100)(I_B) + (99I_B)(1,2k) + v_{CE} + 1k(100)(I_B) = 0$$

$$50\mu$$

$$50\mu$$

$$50\mu$$